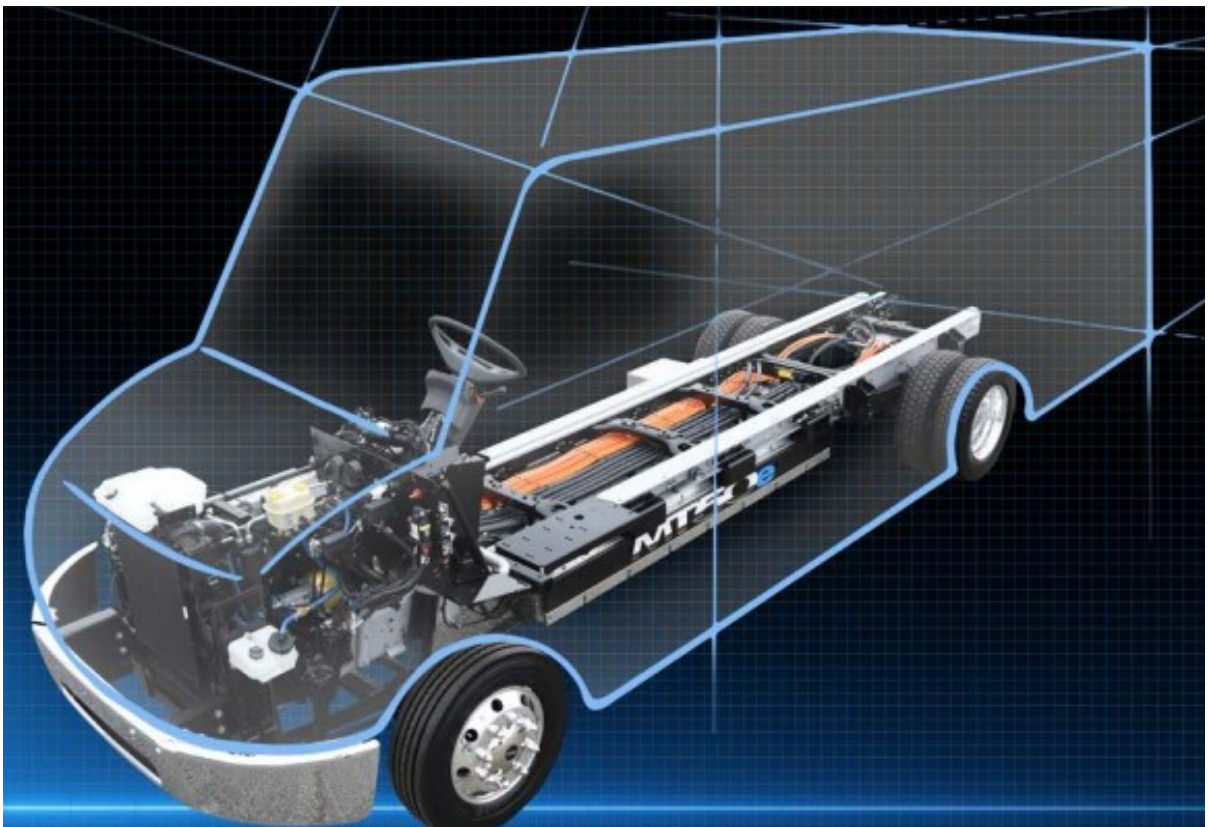




MT50e
CHASSIS

Models 2021 to Present

Emergency Response Guide





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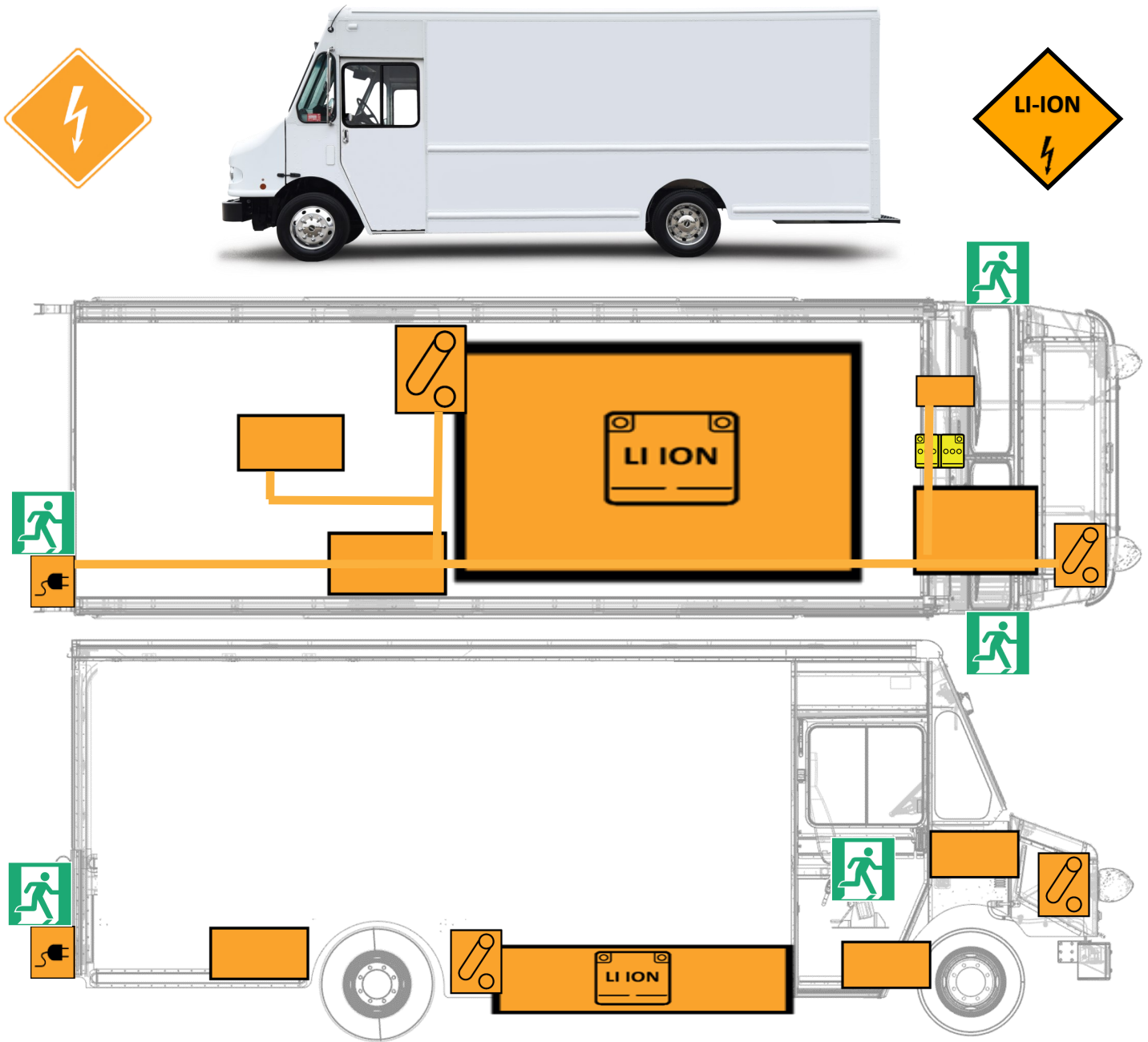
8. Towing / transportation / storage

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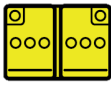


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NOTE: Representative schematic of vehicle exterior is shown. Various door opening points are possible

 Electric Propulsion	 Charging Port	 High Voltage Li-Ion Battery	 High Voltage Component
 Low Voltage Battery	 High Voltage Power Cable	 Access to Occupants	 Disconnect High Voltage



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1. Identification

Vehicle identification methods may vary by final stage manufacturer. Refer to the final stage manufacture documentation for specific guidance on identifying electric vehicle configurations. For applicable models, the high-voltage charge point is located on the passenger side at the rear of the vehicle.

2. Immobilization / stabilization / lifting

3. Apply the parking brake.
4. Chock the wheels to immobilize the vehicle.



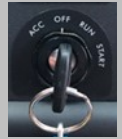


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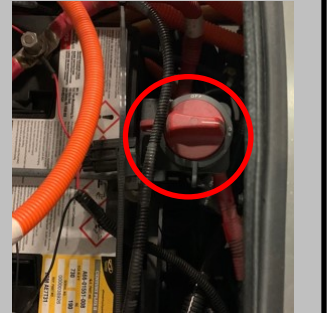
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3. Disable direct hazards / safety regulations

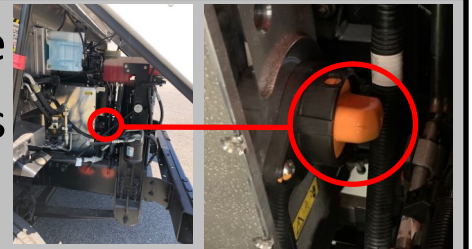
- 1 Turn off and remove the ignition key.



- 2 Turn off the low-voltage battery.
Disconnect all battery ground cables located under the engine access cover in the cab.



- 3 Turn off the HV disconnect on the high voltage junction box, which is located under the hood.



- 4 Remove manual service disconnects (MSDS) to disconnect the high voltage circuit.



(only performed when wearing High Voltage Arc Flash PPE)

- 5 Wait five Minutes for the HV components to discharge.



⚠ WARNING

Always wait five (5) minutes after deactivating the high voltage prior to any work on the vehicle. The ensures the systems high voltage is properly dissipated.



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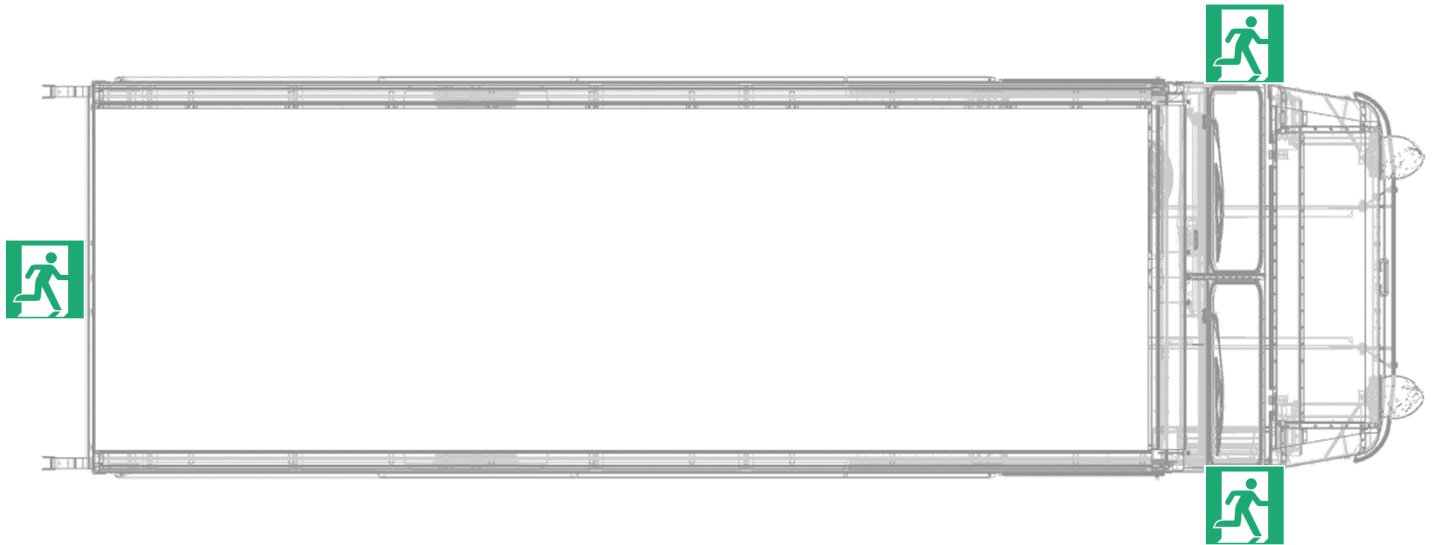
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4. Access to occupants



There are three door exits located in the front of the vehicle on either side.

NOTE: Representative schematic of vehicle exterior is shown. Various door opening points are possible.



WARNING

For extrication purposes, the MT50e is expected to contain at least one and potentially two occupants or more occupants, depending on cab type and other factors. This vehicle contains:

1. High-Voltage (HV) components and cabling primarily routed between the frame rails and beneath the floor; and
2. Charge ports on the right-hand side and routed below the floor level, which are only energized during charging.

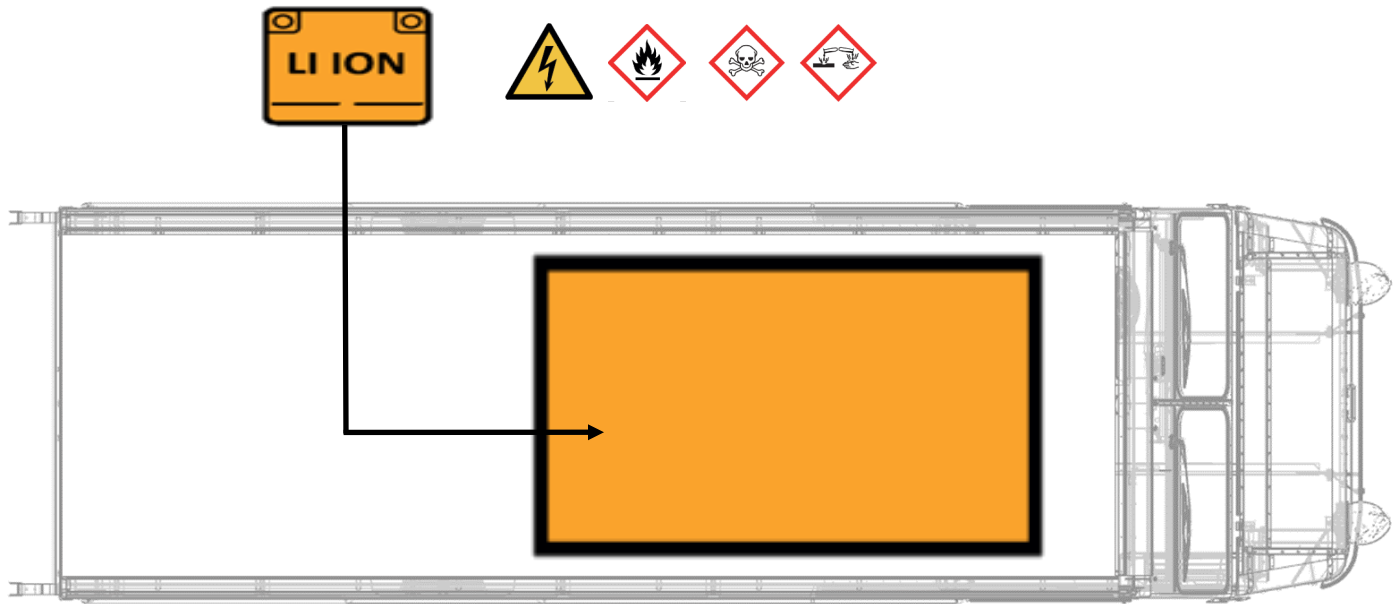
The HV system can be safely disabled using either the primary shut-off switch located on the left side of the battery or the secondary switch positioned under the hood on the right-hand side, as shown in the diagram. Responders should confirm the exact location of all components before cutting or piercing, as the vehicle may have been altered from its original design.



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5. Stored energy / liquids / gases / solids





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6. In case of fire

Lithium-ion batteries contain liquid, flammable electrolyte.

Burning batteries can also ignite other batteries in the vicinity.

The extinguishing agent must be applied continuously until the battery is fully cooled down; otherwise, there is a risk of re-ignition.

- A burning lithium-ion battery generally cannot be extinguished directly. Water can be used as an extinguishing agent for cooling lithium-ion batteries.
- A battery fire may continue to burn for several hours or re-ignite, so it is recommended to cool the battery continuously with excessive amounts of water.
- The temperature of the battery can be monitored with a thermal imaging camera to ensure it is not heating up.
- Re-check the temperature of the battery once an hour to ensure it is not increasing. If the temperature rises, continue to cool the battery with water.

Do not store a vehicle containing a damaged or burning lithium-ion battery within 15 feet of a structure or other vehicles.

WARNING

Always wear appropriate PPE when fighting vehicle fires

For fighting vehicle fires with lithium-ion batteries, use PPE at least as protective as used when fighting conventional vehicle fires, including but not limited to SCBA and full-body protective clothing





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7. In case of submersion

In the recovery of submersion or partly submerged vehicles equipped with a high-voltage system, the handling procedure is the same as for a conventional vehicle. However, a high-voltage system immersed in water may pose an increased risk of electric shock. Until the voltage has been confirmed as dead by a trained professional, appropriate safety measures must be taken.

After the vehicle has been recovered from the water, follow the previously described process for disabling the high-voltage system.

DANGER

Handling a submerged vehicle without appropriate training and personal protective equipment (PPE) can result in serious injury or death.

A submerged vehicle should be handled by emergency personnel while wearing the appropriate PPE.

WARNING

Removal of a submerged service disconnect can result in an electrical short and potential fire leading to serious injury or death.

Submersion in water, especially salt water, can damage both low and high voltage components. Although not a common occurrence, this could result in an electrical short and potential fire once the vehicle is no longer submerged.

WARNING

Damaged high voltage batteries can produce flammable gas and potential fire leading to serious injury or death.

Ventilate the passenger compartment once the vehicle is out of the water. Do not store the vehicle indoors.

The high-voltage system of the MT50e is isolated from the chassis. When undamaged, the system will not energize the surrounding water, even when fully submerged.

Emergency responders will check for damage and, after removing the vehicle from the water, will disable the high-voltage system.



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8. Towing / transportation / storage

Towing (Front Hookup Only)

⚠ DANGER

Be careful of electric shock caused by current flowing to the vehicle if high voltage equipment or cables are damaged.

⚠ WARNING

Do not tow an unbraked vehicle if the combined weight of both vehicles exceeds the sum of the gross axle weight ratings (GAWR) of the towing vehicle. Otherwise, the brake capacity will be inadequate, which could result in personal injury or death.

IMPORTANT

- When it is necessary to tow the vehicle, ensure that the following instructions are closely followed to prevent damage to the vehicle.
- When towing or pushing the vehicle, regardless of the distance or speed, remove both drive axles and axle shafts. Failure to do this when towing the vehicle with the rear wheels on the ground could result in damage to the e-Axle and other parts.
- The vehicle should never be towed from the rear. The gross axle weight rating (GAWR) of the front axle may not be sufficient to support the increased load when towing from the rear, which could damage the front axle.
- Towing rules and regulations vary by federal, state, local, and transit authority. These laws must be followed when towing the van.

1. Block the front and back of at least one of the vehicle's tires to prevent the truck from moving during the procedure.
2. Engage the parking brake.
3. Starting on the left-hand side wheel, place a drip pan under the end of the wheel hub.
4. Remove the axle shaft nuts, washers, and tapered dowels, if used.
5. Remove the axle shaft from the drive axle housing.

CAUTION: Do not use a chisel or any other wedge device to loosen the shaft, as chisels and wedges will damage the flange of the wheel hub.

6. Place a plastic bag over the flanged end of the axle shaft first. Pull the bag over the splined end and fold it back down. Secure the bag in place with a couple of wraps of electrical tape around the shaft.
7. Return to the wheel end and remove the axle shaft gasket. Discard the gasket.



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8. Wipe the end of the hub to remove any oil.
9. Install a wheel end cover.
10. Reinstall the wheel end fasteners and tighten in a criss-cross pattern. Do not overtighten.
11. Go to the right side of the rear-rear axle and repeat the process

Storage of Damaged Battery

1. If the lithium-ion battery has been damaged, it may increase in temperature and lead to a fire. Use a thermal imaging camera to ensure the battery temperature does not exceed 60°C. If necessary, initiate cooling with water.
2. Before handling the damaged battery, ensure there is no smoke or signs of heat. If, after observing the battery pack, there are no signs of heat and the high-voltage system has been disabled, the battery may be moved to a safe location.
3. Ensure that the damaged battery or vehicle with a damaged battery has a 15-foot buffer area around it, away from buildings or materials.